Optimal Provable Robustness of Quantum Classification via Quantum Hypothesis Testing

Maurice Weber*, Nana Liu†, Bo Li‡, Ce Zhang* & Zhikuan Zhao*

*ETH Zürich, †Shanghai Jiao Tong University, ‡University of Illinois Urbana-Champaign

Introduction

Quantum classification algorithms are at the intersection between quantum computation and machine learning and are recognized as having the potential to offer quantum speedup or superior predictive accuracy. However, they have been shown to be just as vulnerable to input perturbations [1] like their classical counter-parts [2]. These perturbations can occur either due to imperfect implementation, prevalent in the NISQ era, or, more menacingly, due to adversarial attacks where a malicious party aims to fool a classifier by crafting practically undetectable noise patterns which trick a model into misclassifying a given input. Here, we aim to address these short-comings and derive conditions under which classifiers are guaranteed to be robust. Specifically, we establish a fundamental connection between quantum hypothesis testing and the robustness of quantum classifiers against unknown noise sources. This naturally leads to a robustness condition which is tight under modest assumptions.

Quantum classification

A K-class quantum classifier is a map $\mathcal{A}\colon \mathcal{S}(\mathcal{H}) \to \mathcal{C}$, mapping density operators to class labels $k \in \mathcal{C}$, $|\mathcal{C}| = K$. Given a POVM $\{\Pi_k\}_k$ and a CPTP map $\mathcal{E}$, the classification output is defined to be

$$\mathcal{A}(\sigma) = \arg\max_k \operatorname{Tr}\left[\Pi_k \mathcal{E}(\sigma)\right]$$

which is the most likely outcome of the measurement performed on the evolved state $\mathcal{E}(\sigma)$.

Hypothesis Testing

Quantum Hypothesis Testing (QHT) is concerned with the question of how to optimally discriminate between quantum states σ and ρ. Here, we consider asymmetric hypothesis testing, where the goal is to minimize the probability of conducting a type-II error under the constraint that the type-I error is no larger than some predefined value. Formally, we aim to solve

$$\begin{aligned} \beta^*_\alpha(\rho, \sigma) = \quad & \text{minimize} \quad \operatorname{Tr}\left[\rho(\mathbb{1} - M)\right] \\ & \text{s.t.} \quad \operatorname{Tr}\left[\sigma M\right] \leq \alpha, \\ & \qquad\; 0 \leq M \leq \mathbb{1}. \end{aligned}$$

Threat model

An adversarial attack is modeled by a CPTP map $\mathcal{E}_A\colon \mathcal{S}(\mathcal{H}) \to \mathcal{S}(\mathcal{H})$. The adversary applies the transform $\sigma \to \rho := \mathcal{E}_A(\sigma)$ with the goal of flipping the prediction, i.e. $\mathcal{A}(\rho) \neq \mathcal{A}(\sigma)$.

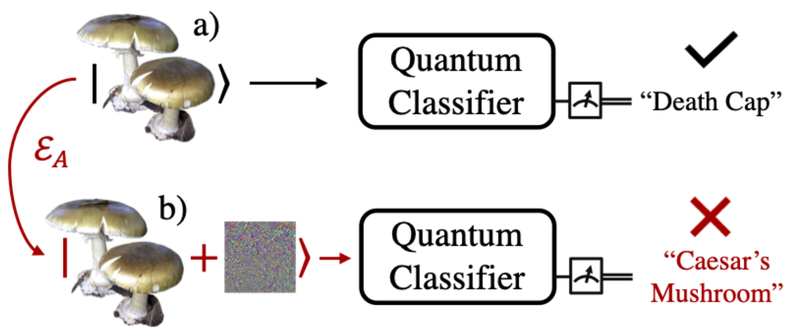

Figure 1: a) A quantum classifier correctly classifies the (highly toxic) mushroom as "death cap". b) An adversary perturbs the image to fool the classifier into believing it to be the edible "Caesar's mushroom".

Main Result

Theorem 1. *Let $\sigma, \rho \in \mathcal{S}(\mathcal{H})$ be benign and adversarial quantum states and let $\mathcal{A}$ be a quantum classifier. Suppose that for $k_A \in \mathcal{C}$ and $p_A, p_B \in [0, 1]$ the classifier $\mathcal{A}$ has class probabilities which satisfy*

$$\operatorname{Tr}\left[\Pi_{k_A}\mathcal{E}(\sigma)\right] \geq p_A > p_B \geq \max_{k \neq k_A} \operatorname{Tr}\left[\Pi_k \mathcal{E}(\sigma)\right]. \tag{1}$$

Then, it is guaranteed that $\mathcal{A}(\rho) = \mathcal{A}(\sigma)$ for any ρ which satisfies

$$\beta^*_{1-p_A}(\rho, \sigma) + \beta^*_{p_B}(\rho, \sigma) > 1. \tag{2}$$

Under modest assumptions, this robustness condition is provably tight in the sense that, whenever the adversarial state ρ violates this condition, there is a classifier $\mathcal{A}^\star$ which is consistent with the class probabilities but which classifies ρ differently from σ.

Theorem 2. *Suppose $p_A + p_B = 1$. If the adversarial state ρ violates condition (2), there exists a quantum classifier $\mathcal{A}^\star$ that is consistent with the class probabilities (1) and for which $\mathcal{A}^\star(\rho) \neq \mathcal{A}^\star(\sigma)$.*

The additional assumption covers binary classification in full generality and multi-class classification whenever the most likey class is measured with probability larger than 1/2.

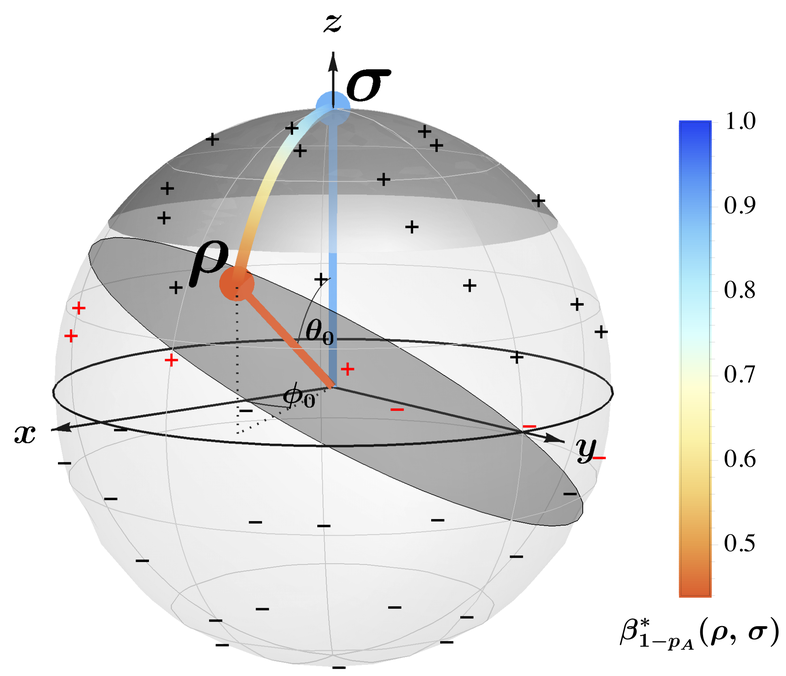

Figure 2: Robustness for an example classifier (grey disk) which classifies σ and ρ differently. Indeed, the adversarial state is outside the robust region (spherical cap). At the same time, the optimal type-II error probability β^* for distinguishing ρ from σ is smaller than 1/2, in violation of our robustness condition.

Robustness in terms of trace distance

Pure states

If both ρ and σ are pure and $p_A + p_B = 1$, the robustness condition (2) can be equivalently expressed in terms of trace distance. Namely, if ρ satisfies

$$\frac{1}{2}\left\|\rho - \sigma\right\|_1 < \sqrt{\frac{1}{2} - \sqrt{p_A(1 - p_A)}}$$

then it is guaranteed that $\mathcal{A}(\rho) = \mathcal{A}(\sigma)$. The general case where $p_A + p_B < 1$ has a similar closed form solution.

Mixed states

For mixed states, a robustness bound can be derived from Hölder duality. Specifically, if ρ satisfies

$$\frac{1}{2}\left\|\rho - \sigma\right\|_1 < \frac{p_A - p_B}{2}$$

then it is guaranteed that $\mathcal{A}(\rho) = \mathcal{A}(\sigma)$. However, while this bound is sufficient, it is not necessary in general.

Conclusions

- We have seen how QHT can be leveraged to obtain robustness guarantees for classification algorithms.
- This robustness guarantee is provably optimal under modest assumptions.
- Based on these results, we can develop protocols that allow us to rigorously reason about the resilience of quantum classifiers against adversaries and, more generally, against any unknown noise source.

References

[1] N. Liu and P. Wittek. Vulnerability of quantum classification to adversarial perturbations. *Phys. Rev. A*, 101:062331, Jun 2020.

[2] I. Goodfellow, J. Shlens, and C. Szegedy. Explaining and harnessing adversarial examples. In *International Conference on Learning Representations*, 2015.